MIL-E-46495A (AR)
AMENDMENT 2
21 NOVEMBER 1980

SUPERSEDING
AMENDMENT 1
5 April 1972

MILITARY SPECIFICATION
EXPLOSIVE COMPOSITION HTA-3

This amendment forms a part of Military Specification MIL-E-46495 (AR) dated 30 November 1964.

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Paragraph 3.2: Delete TABLE II in its entirety and substitute:

* "TABLE II

Material	Percent Type I	Percent Type II	Applicable Paragraphs
TNT	29.0 ± 2.0	28.65 ± 2.0	4.3.1.1
HMX	49.0 ± 2.0	49.0 ± 2.0	4.3.1.2
Aluminum	22.0 ± 2.0	22.0 ± 2.0	4.3.1.4
Calcium Silicate	----------	0.35 ± 0.15	4.3.1.3"

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Paragraph 4.2.3: Add a new paragraph as follows:

* "4.2.3 Testing

PRECAUTION WARNING

This specification covers sampling and testing of toxic or hazardous materials. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials."

FSC: 1375

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Paragraph 4.3.1: Delete in its entirety and substitute:

* "4.3.1 Composition.

4.3.1.1 TNT. Grind the sample to a fine powder using remote grinding equipment and observing proper safety precautions.

Store ground sample in a conductive rubber container. Transfer an accurately weighed portion of approximately 2.5 grams into a tared 50 ml high form medium porosity filtering crucible (Weigh to the nearest 0.0001 gram). Extract with 30-40 ml portions of toluene saturated with HMX until a total volume of at least 250 ml has been used. Allow each portion to soak the crucible contents for approximately five (5) minutes. (The completeness of the TNT extraction may be checked by testing the filtrate with Webster's Reagent. A dark pink color indicates TNT is present). Draw air through the crucible until it appears dry. Dry the crucible and contents in an oven maintained at 100±5 degrees Centigrade (^{o}C) for one (1) hour. Cool the crucible and contents in a desiccator and weigh. Retain the crucible and contents for the determinations which follow.

Calculate the TNT content of the sample as follows:

$$\text{Percent TNT} = \frac{100\ ((A-B)-MW)}{W-(MW)}$$

Where: A = Weight of crucible and sample, g
B = Weight of crucible and residue, g
W = Weight of sample g
M = Moisture content in the sample expressed as a decimal (4.3.2)

4.3.1.2 HMX. Place the crucible and contents retained from the TNT determination on a filtering apparatus and wash with ten 20 ml portions of acetone. Allow each portion of the acetone to remain in contact with the residue in the crucible for one minute before applying suction. After visible HMX has been extracted, place the crucible in a 150 ml beaker, add about one inch of acetone to the crucible and the beaker, and place the beaker in an ultrasonic cleaner for one-half hour. Remove the crucible, aspirate the remaining acetone through the filter and wash the crucible several times with acetone. Aspirate the crucible and contents until the odor of acetone is no longer detectable. Dry the crucible and contents in an oven maintained at 100±5°C for one hour. Cool in a desiccator and weigh.

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Calculate the HMX content of the sample as follows:

$$\text{Percent HMX} = \frac{100\ (A-B)}{W-(MW)}$$

Where: A = Weight of residue in crucible retained from 4.3.1.1
B = Weight of residue in crucible, g
W = Weight of sample, g
M = Moisture content, expressed as a decimal (4.3.2)

4.3.1.3 Calcium silicate. Wash the contents of the crucible remaining after the HMX determination into a 150 ml teflon or heat-resistant plastic beaker with 50-75 ml of distilled water (a plastic squeeze bottle works best). Add 40 ml of 20% (by weight) sodium hydroxide solution to the beaker and cover with a teflon (or plastic) cover.

Add 10 ml of 20% sodium hydroxide solution to the crucible and allow the solution to react with aluminum remaining on the fritted glass and crucible wall until the reaction is complete and the crucible appears clear of aluminum residue. Set the crucible in a pan during the reaction and rotate the crucible to react residues on the crucible wall as necessary. The sodium hydroxide solution should remain in the crucible no more than 15 minutes or the crucible will be etched excessively.

After the reaction is complete, transfer the crucible contents into a teflon beaker with a distilled water rinse. Then immediately extract the crucible by vacuum and wash several times with hot distilled water. After rinsing, set the crucible aside until final filtration.

After the reaction of the crucible contents in the teflon beaker subsides (about 20 minutes), heat the beaker and contents on a steam plate until all gray-black material has reacted (about 15 minutes) and the solution with residue is a clear to rusty color. Inspect the beaker to insure that all material is rinsed off the beaker walls into the solution.

After the reaction in the beaker is complete and while still hot, quantitatively transfer and filter the contents through the crucible with distilled water. (CAUTION: Do not allow the beaker contents to set more than one hour after sodium hydroxide is added as an insoluble compound may be formed). Immediately rinse the crucible and contents several times with distilled water followed by several acetone rinses.

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Aspirate to dryness and dry the crucible and contents in an oven maintained at $100^{o}\pm5^{o}C$ for one hour, cool in a desiccator and weigh.

After weighing, clean the crucible by backflushing with cleaning solution (sulfuric acid with potassium dichromate), hot distilled water and acetone. Dry in $100^{o}\pm5^{o}C$ oven for one hour, cool in a desiccator and reweigh. (This weight is required to check if crucible tare was altered by sodium hydroxide etching.)

Calculate the calcium silicate content of the sample as follows:

Weight of Calcium Silicate (C) = $W_1 - W_2$

Where: W_1 = Weight of crucible and contents after aluminum extraction
W_2 = Weight of crucible after cleaning

Percent Calcium Silicate = $\frac{100\ C}{W - (MW)}$

Where: C = Weight of calcium silicate, g
W = Weight of sample g
M = Moisture content, expressed as a decimal

4.3.1.4 Aluminum. Calculate the aluminum content as follows:

Percent Aluminum = 100 - (A plus B plus C)

Where: A = Percent TNT (4.3.1.1)
B = Percent HMX (4.3.1.2)
C = Percent Calcium Silicate (4.3.1.3)"

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Paragraph 4.3.2.1: Delete in its entirety and substitute:

* "4.3.2.1 Special solvent. The special solvent used in the determination of moisture shall consist of equal volumes of anhydrous methanol and toluene thoroughly mixed. If necessary the solvents shall be dried by distillation."

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